UTERINE FIBROID

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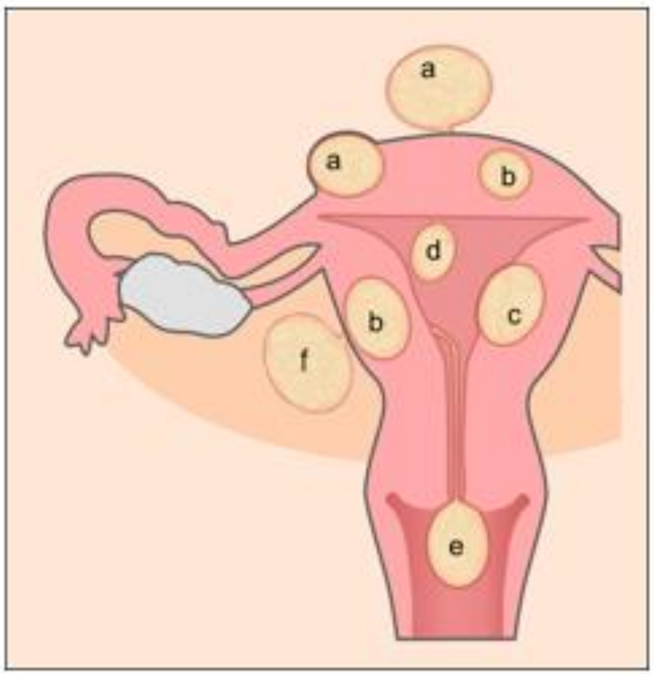
- A fibroid is a benign tumour of uterine smooth muscle, termed a leiomyoma.
- The gross appearance is of a firm, whorled tumour located adjacent to and bulging into the endometrial cavity (submucous fibroid), centrally within the myometrium (intramural fibroid), at the outer border of the myometrium (subserosal fibroid) or attached to the uterus by a narrow pedicle containing blood vessels (pedunculated fibroid).
- Fibroids can arise separately from the uterus, especially in the broad ligament, presumably from embryonal remnants. The typical whorled appearance may be altered following degeneration, three forms of which are recognized: red, hyaline and cystic.

A



a = subserosal fibroids
b = intramural fibroid
c = submucosal fibroid
d= pedunculated submucosal fibroid
e = fibroid in statu nascendi
f = fibroid of the broad ligament

B


Multiple uterine fibroids of variable sizes studded on all over uterus
Rajshree D. Katke

Loss of worled appearance if degenration occurred

- Red degeneration follows an acute disruption of the blood supply to the fibroid during active growth, classically during pregnancy. This may present with the sudden onset of pain and tenderness localized to an area of the uterus, associated with a mild pyrexia and leukocytosis.
- The symptoms and signs typically resolve over a few days and surgical intervention is rarely required.
- Hyaline degeneration occurs when the fibroid more gradually outgrows its blood supply, and may progress to central necrosis, leaving cystic spaces at the center, termed cystic degeneration.
- As the final stage in the natural history, calcification of a fibroid may be detected incidentally on an abdominal X-ray in a postmenopausal woman.

- Rarely, malignant or sarcomatous degeneration has been said to occur, but malignancy probably arises through a separate pathway of chromosomal deletions and the real possibility of malignant change in a fibroid is vanishingly small.
- Clinical features Fibroids are common, being detectable clinically in about 20 per cent of women over 30 years of age. Autopsy studies with systematic histology of the uterus show a prevalence of up to 50 per cent.
- Risk factors for clinically significant fibroids are nulliparity, obesity, a positive family history and African racial origin. The great majority do not cause symptoms but may be identified coincidentally, for example at the time of taking a cervical smear or performing laparoscopic sterilization.

Symptoms and signs

- menstrual disturbance and pressure symptoms, especially urinary frequency.
- Pain is unusual except in the special circumstance of acute degeneration discussed above.
- Menorrhagia may occur coincidentally in a woman with fibroids; it is likely that only submucous fibroids distorting the endometrial cavity and increasing the surface area

- Subfertility may result from mechanical distortion or occlusion of the Fallopian tubes, and an endometrial cavity grossly distorted by submucous fibroids may prevent implantation of a fertilized ovum.
- Once a pregnancy is established, however, the risk of miscarriage is not increased. In late pregnancy, fibroids located in the cervix or lower uterine segment may be the cause of an abnormal lie.
- After delivery, postpartum hemorrhage may occur due to inefficient uterine contraction.
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- Abdominal examination might indicate the presence of a firm mass arising from the pelvis, and on bimanual examination the mass is felt to be part of the uterus, usually with some mobility.
- Differential diagnosis Other causes of an abdominopelvic mass in a woman in the reproductive years need to be considered.
- The uterus enlarged with fibroids is typically firm in contrast to a uterus enlarged with a pregnancy.
- An ovarian tumour, whether benign or malignant, primary or secondary, may enlarge to occupy the pelvis and be clinically difficult to differentiate from a uterine fibroid.
- Leiomyosarcomas typically present with a history of a rapidly enlarging abdominopelvic mass. There may be less mobility of the uterus than expected with a fibroid and general signs of cachexia

Investigation

- Often the clinical features alone will be sufficient to establish the diagnosis.
- A hemoglobin concentration will help to indicate anemia if there is clinically significant menorrhagia. Ultrasonography is useful to distinguish a uterine from an ovarian mass.
- Imaging of the renal tract may be helpful in the presence of a large fibroid to exclude hydronephrosis due to pressure from the mass on the ureters.
- Clinical suspicion of sarcoma will be an indication for needle biopsy or, more likely, urgent laparotomy.

Treatment

- Conservative management is appropriate where asymptomatic fibroids are detected incidentally.
- It may be useful to establish the growth rate of the fibroids by repeat clinical examination or ultrasound after a 6–12- month interval.
- Where treatment is required, the only practical currently available medical treatment is ovarian suppression using a gonadotrophin-releasing hormone (GnRH) agonist. Unfortunately, while very effective in shrinking fibroids, when ovarian function returns, the fibroids regrow to their previous dimensions.
- Mifepristone (an antiprogestogen) has been shown to be effective in shrinking fibroids at a low dose, but is not available for use in this indication. The optimal dose, duration of treatment and long-term effects have yet to be established

- The choice of surgical treatment is determined by the presenting complaint and the patient’s aspirations for menstrual function and fertility. Menorrhagia associated with a submucous fibroid or fibroid polyp may be treated by hysteroscopic resection.
- Where a bulky fibroid uterus causes pressure symptoms, the options are myomectomy with uterine conservation, or hysterectomy. Myomectomy will be the preferred option where preservation of fertility is required,

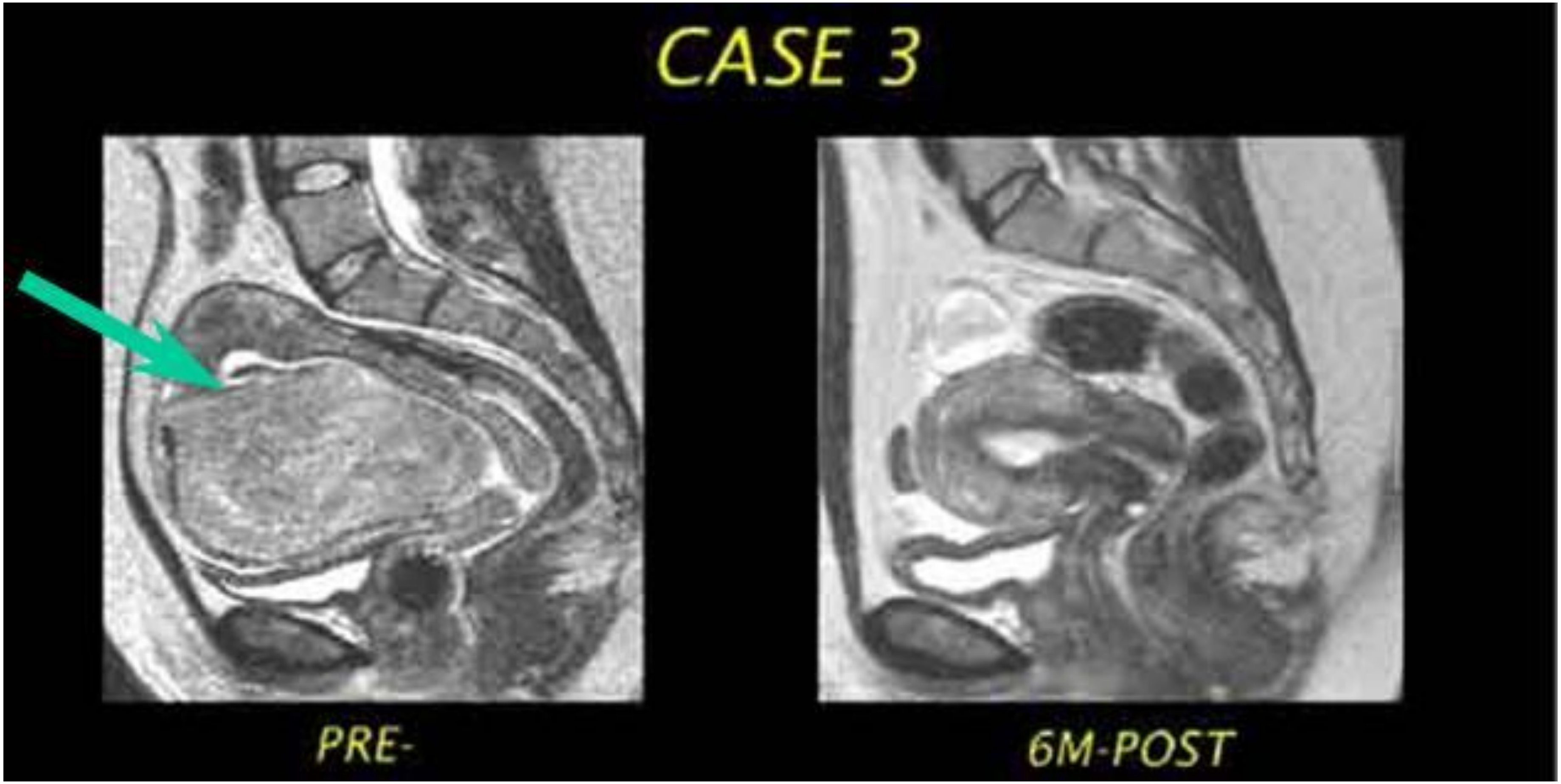
- , but care must be taken in the management of a subsequent pregnancy, as the uterus may be predisposed to rupture. It is traditionally held that uterine rupture during pregnancy is more likely when the endometrial cavity has been entered during myomectomy, but, not surprisingly, there are few data to confirm or refute this.
- In any event, the decision to undertake myomectomy in a woman who desires future fertility needs to be carefully considered and the benefits and risks fully discussed with the patient.
- An important point for the preoperative discussion is that there is a small but significant risk of uncontrolled bleeding during myomectomy, which could lead to the need for hysterectomy

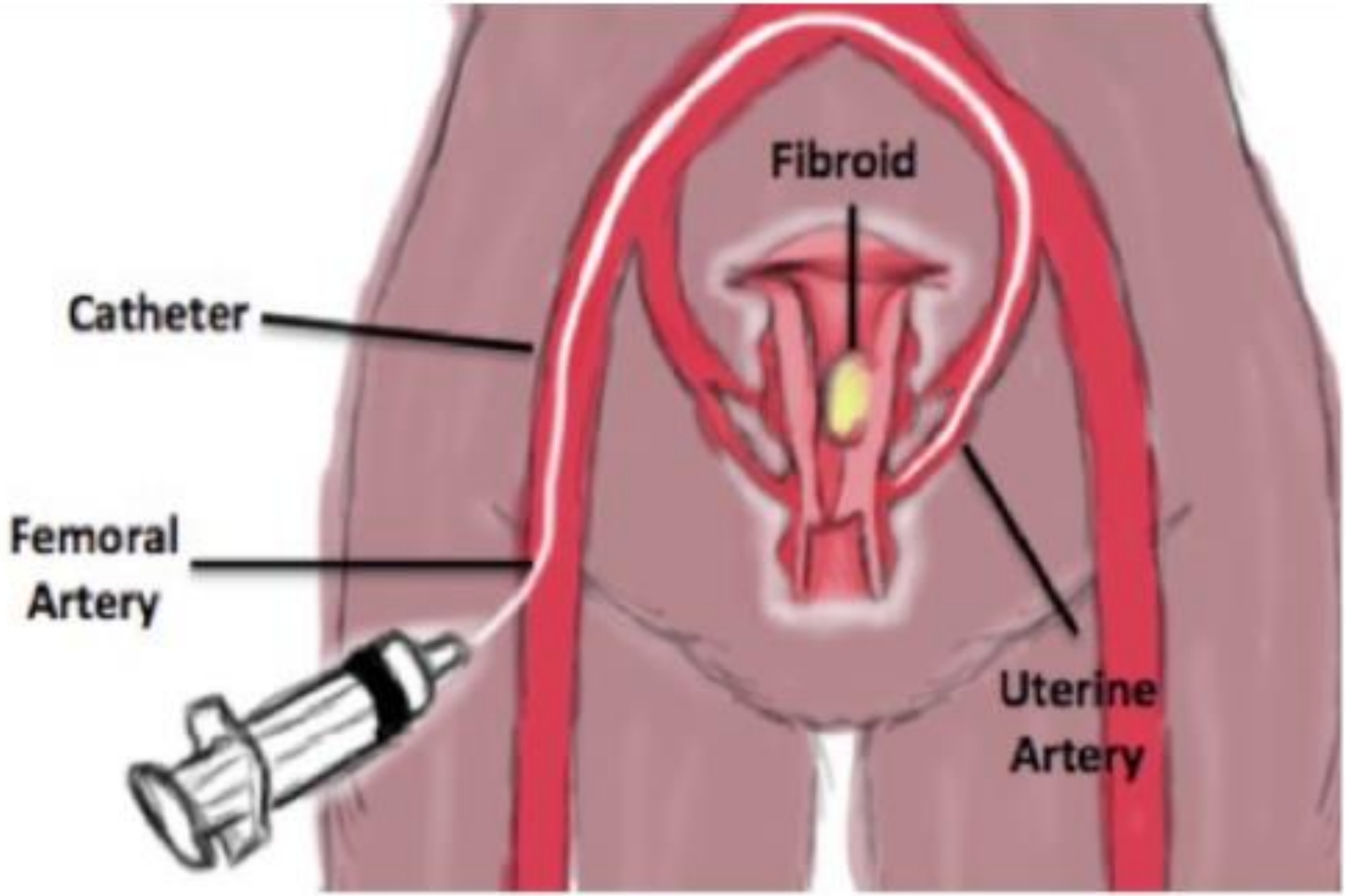
- hysterectomy and myomectomy can be facilitated by GnRH agonist pretreatment over a 2-month period to reduce the bulk and vascularity of the fibroids.
- Useful benefits of this approach are to enable a Pfannensteil (low transverse) rather than a midline abdominal incision, or to facilitate vaginal rather than abdominal hysterectomy, both of which are conducive to more rapid recovery and fewer postoperative complications.
- A technical problem with myomectomy after GnRH agonist pre-treatment is that the tissue planes around the fibroid are less easily defined, but on the positive side, blood loss and the likely need for transfusion are reduced.

New development

- Endoscopic surgical treatments for fibroids have proved disappointing: myolysis using a diathermy needle to destroy the tissue is followed by intense adhesion formation.
- Given the requirement for a substantial blood supply to support growth, interruption of the arterial supply to the tumour is a theoretically attractive concept. In practice, this is feasible by the radiological technique of percutaneous selective catheterization of the uterine arteries. Microparticles are released into the vessels, causing occlusion of both uterine arteries.
- Sufficient collateral circulation is present from the ovarian arteries to sustain normal uterine metabolic requirements, and women experience a substantial reduction in fibroid bulk, together with improvement in menstrual symptoms over the following 6 months. Currently available follow-up data suggest that the symptomatic improvement is sustained.

- Pre and post embolization




Fibroid
Catheter
Femoral
Artery
Uterine
Artery

- Uterine fibroid embolisation, MRI-guided focused ultrasound ablation and laparoscopic uterine artery occlusion are grouped under a category called minimally invasive uterus preserving treatments, which are considered alternatives to abdominal myomectomy/hysterectomy

Treatment hierarchy	Seeking contraception	Wishing to conceive
First step	COCs, Oral progestogens,I njected progestogens ,Mirena LNG-IUS,Short (<6 month) course of GnRHa	Tranexamic acid, NSAIDs
Second step	**Hysteroscopic myomectomy** +/- Endometrial resection/ablation +/- Mirena LNG-IUS	**Hysteroscopic myomectomy,** **Laparoscopic myomectomy**
Second step (minimally invasive uterus-conserving treatments)	Uterine artery embolisation Magnetic resonance-guided focused ultrasonography Laparoscopic uterine artery occlusion with/without utero-ovarian occlusion Transvaginal Doppler guided uterine artery occlusion **Bipolar radiofrequency ablation** (intrauterine ultrasound-guided or laparoscopic-guided)	
Third step	Hysterectomy +/- bilateral salpingo-oophorectomy	**Abdominal myomectomy**

Thank you